```python
import numpy as np

import matplotlib.pyplot as plt

import pandas as pd

import numpy as np

dataset = np.array([
    ['Asset Flip', 100, 1000],
    ['Text Based', 500, 3000],
    ['Visual Novel', 1500, 5000],
    ['2D Pixel Art', 3500, 8000],
    ['2D Vector Art', 5000, 6500],
    ['Strategy', 6000, 7000],
    ['First Person Shooter', 8000, 15000],
    ['Simulator', 9500, 20000],
    ['Racing', 12000, 21000],
    ['RPG', 14000, 25000],
    ['Sandbox', 15500, 27000],
    ['Open-world', 16500, 30000],
    ['MMOFPS', 25800, 52000],
    ['MMORPG', 30000, 80000]
])

print(dataset)
```

```
[['Asset Flip' '100' '1000']
 ['Text Based' '500' '3000']
 ['Visual Novel' '1500' '5000']
 ['2D Pixel Art' '3500' '8000']
 ['2D Vector Art' '5000' '6500']
 ['Strategy' '6000' '7000']
 ['First Person Shooter' '8000' '15000']
 ['Simulator' '9500' '20000']
 ['Racing' '12000' '21000']
 ['RPG' '14000' '25000']
 ['Sandbox' '15500' '27000']
 ['Open-world' '16500' '30000']
 ['MMOFPS' '25800' '52000']
 ['MMORPG' '30000' '80000']]
```

```python
X = dataset[:, 1:2].astype(int)
print(X)
```

```
[[  100]
 [  500]
 [ 1500]
 [ 3500]
 [ 5000]
 [ 6000]
```

```
 [ 8000]
 [ 9500]
 [12000]
 [14000]
 [15500]
 [16500]
 [25800]
 [30000]]
```

```
y = dataset [:, 2].astype (int)
print(y)
```

```
[ 1000  3000  5000  8000  6500  7000 15000 20000 21000 25000 27000
30000
 52000 80000]
```

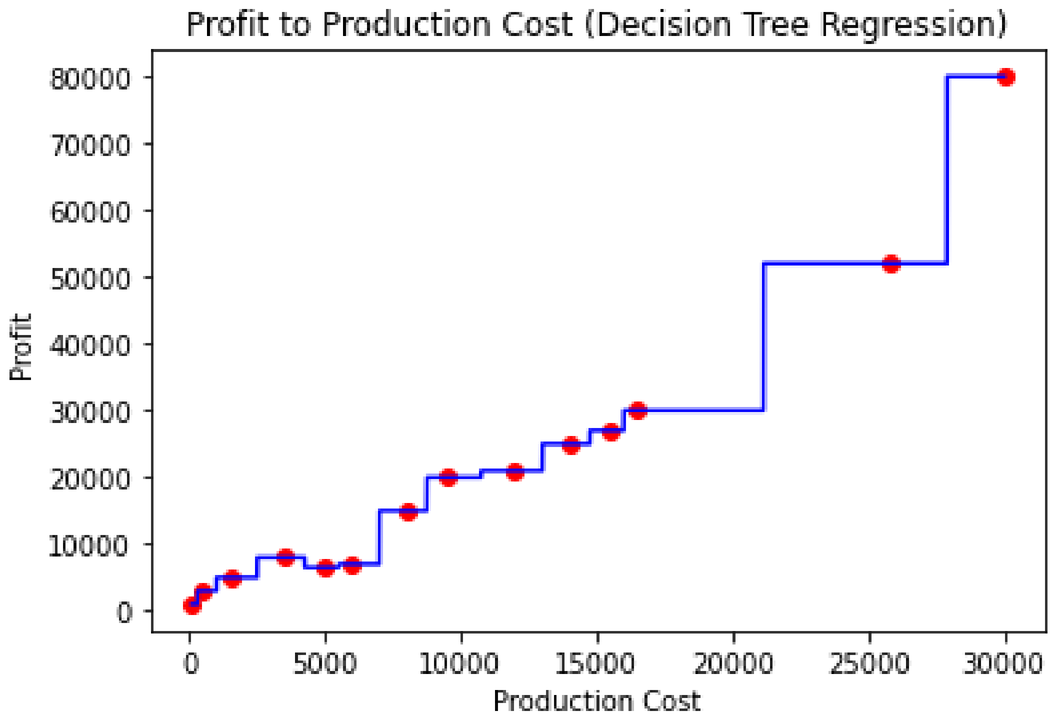
```
from sklearn.tree import DecisionTreeRegressor
regressor = DecisionTreeRegressor (random_state = 0)
regressor.fit(X, y)
```

```
DecisionTreeRegressor(random_state=0)
```

```
y_pred = regressor.predict([[3750]])
print("Predicted price: %d\n"%y_pred)
```

```
Predicted price: 8000
```

```
X_grid = np.arange(min(X), max(X), 0.01)
X_grid = X_grid.reshape((len(X_grid), 1))
plt.scatter (X, y, color = 'red')
plt.plot(X_grid, regressor.predict(X_grid), color = 'blue')
plt.title('Profit to Production Cost (Decision Tree Regression)')
plt.xlabel('Production Cost')
plt.ylabel('Profit')
plt.show()
```


Profit to Production Cost (Decision Tree Regression)
80000
70000
60000
50000
40000
30000
20000
10000
0
Profit
0
5000
10000
15000
20000
25000
30000
Production Cost